ELEC70006
EE9-CS1-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Tuesday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : I. Jaimoukha
Second Marker(s) : E. Kerrigan

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1. Consider the feedback system in Figure 1. Here, $G(s)$ is a nominal system model and $K(s)$ is a controller. The transfer matrix $\Delta_m(s)$ denotes a stable uncertainty. The design specifications are as follows:

   i) The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable when $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|$, for all ω, where $w_m(s)$ is an appropriate weighting function.

   ii) The performance specifications are to design an internally stabilizing controller such that when $\Delta_m = 0$, the closed-loop should satisfy:

$$\|u(j\omega)\| < |w_u(j\omega)^{-1}| \|d(j\omega)\|,$$

   for all ω, where $w_u(s)$ is an appropriate weighting function.

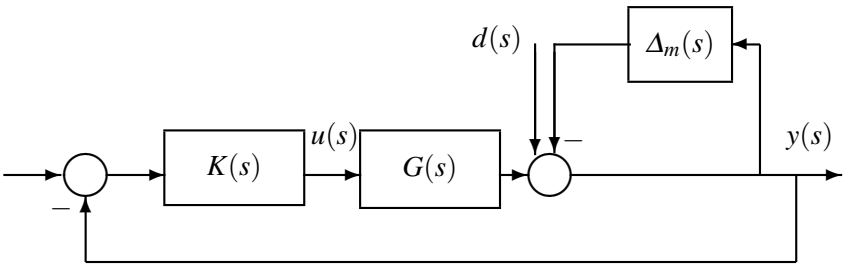


Figure 1

a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix Δ and a feedback transfer matrix H. [5]

b) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_m(s)$ and $w_u(s)$ that are sufficient to achieve the specifications. [5]

c) Define appropriate cost, external, control and measured signals and draw a block diagram, showing these signals, the nominal system model, the controller, and weighting functions. [5]

d) Derive and draw a generalised regulator problem that captures these sufficient conditions. [5]

2. a) Consider a state–variable model described by the dynamic equation

$$\begin{bmatrix} \dot{x}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

and let $H(s) = D + C(sI - A)^{-1}B$ be the transfer matrix. Let

$$M = \begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix},$$

with $P = P^T$ and $\gamma > 0$.

i) State and prove LMI conditions involving A and P that guarantee that A is stable. [4]

ii) State and prove LMI conditions involving P and M that guarantee that A is stable and $\|H\|_\infty < \gamma$. [4]

iii) State dual equivalent conditions involving A and H for the stability of A and for $\|H\|_\infty < \gamma$. Hence state dual LMI sufficient conditions for the stability of A and for $\|H\|_\infty < \gamma$ involving P and another matrix N of a similar form to M. [4]

b) Consider the feedback system in Figure 2. Let $T_{yd}(s)$ denote the transfer matrix from d to y. An internally stabilizing gain matrix F is to be designed such that, for a given $\gamma > 0$, $\|T_{yd}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{yd}(s)$. [4]

ii) Derive sufficient conditions for the existence of a feasible F. Your conditions should be in the form of the existence of solutions to linear matrix inequalities. [4]

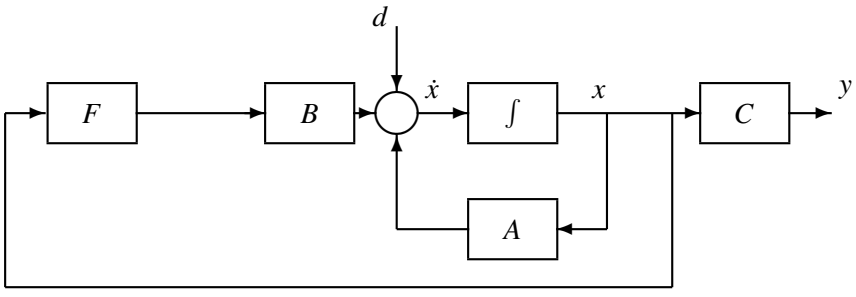


Figure 2

3. Consider the regulator in Figure 3 for which it is assumed that B has full column rank, the pair (A,B) is controllable and $x(0)=x_0$. A stabilizing state-feedback gain matrix F is to be designed such that the cost function

$$J = \|u\|_2^2 + \|x\|_2^2$$

is minimized.

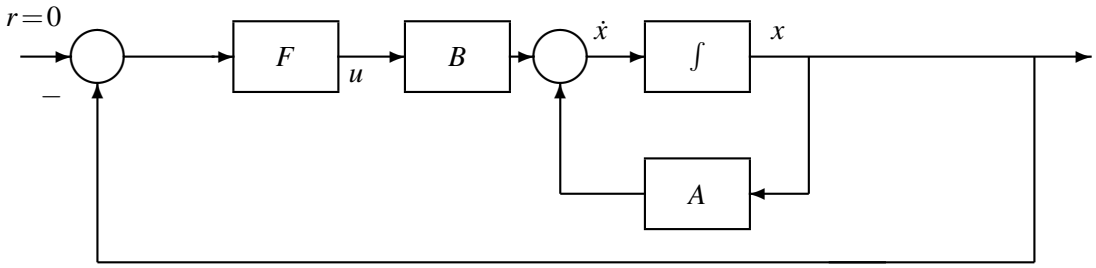


Figure 3

a) By defining suitable cost and Lyapunov functions and completing a square, find an expression for F that achieves the design specifications in terms of the solution of an algebraic Riccati equation. Give the minimum value of the cost function as well as the algebraic Riccati equation. [10]

b) Prove that, for F chosen in part (a), the closed loop system in Figure 3 is stable. State clearly the assumptions required to guarantee stability. [5]

c) Prove that, for F chosen in part (a), the loop gain in Figure 3 is minimum-phase. [5]

4. Consider the regulator shown in Figure 4 for which it is assumed that the pair (A,B_2) is controllable and the triple (A,B_1,C_1) is minimal. Let $z = [\ z_1^T\ \ z_2^T\ ]^T$. It is required to regulate z in response to the external finite energy disturbance w and the initial state $x(0) = x_0$.

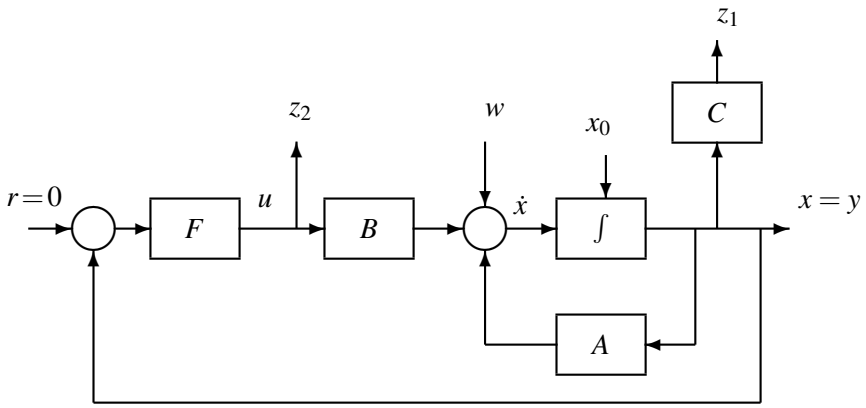


Figure 4

a) For this part, take $w = 0$. A stabilizing state-feedback gain matrix F is to be designed such that $J_0 := \|z\|_2^2$ is minimized.

i) Define a suitable Lyapunov function $V(t)$ in terms of an auxilliary variable P. Obtain an expression for $\int_0^\infty \dot{V}(t)dt$. State clearly any assumptions made. [5]

ii) Find an expression for F that minimizes J_0. Give also the algebraic Riccati equation that needs to be satisfied. [5]

b) For this part, take $x_0 = 0$. Let $H(s)$ denote the transfer matrix from w to z and let $\gamma > 0$ be given. A stabilizing state-feedback gain matrix F is to be designed such that $\|H\|_\infty < \gamma$ for a given $\gamma > 0$.

i) By defining a Lyapunov function and a cost function J involving an auxiliary variable X, and carrying out two completions of squares, derive an expression for J that can be used to solve the problem. [5]

ii) Use the expression for J to formulate the design problem and derive sufficient conditions for its solution. Your conditions should be in the form of the existence of a solution to an algebraic Riccati equation. It should also include an expression for F and an expression for the worst–case disturbance w^*. [5]